Sample Survey

Survey terminology, sampling and non-sampling errors, confidence intervals, and probability and non-probability sampling methods.

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1 Key terms

Census	Complete enumeration of every unit
Sampling unit	Basic unit selected
Sampling frame	Complete list identifying the units
Sampling design	Procedure used to select the sample
Estimator / estimate	Statistic used for a parameter / its particular value
Unbiased	$E(t) = \theta$; e.g. $E(\bar{X}) = \mu$

Sampling fraction

$$f = \frac{n}{N}$$

2 Errors in surveys

Sampling error

Arises from observing a sample, not the whole population. **Absent in a census**. Larger n generally reduces standard error.

Non-sampling error

Everything else: non-response, measurement, recording, processing. Present even in a **census**.

Reliable results need good design, representative sampling, trained enumerators and careful execution.

3 Confidence interval

An **interval estimate** of a parameter.

Confidence level

$$100(1-\alpha)\%$$

$\alpha = 0.05$ → 95%; $\alpha = 0.01$ → 99%. "95%" describes the long-run performance of the method.

4 Probability sampling methods

Method	How	Best when
Simple random (SRSWR / SRSWOR)	Equal inclusion probabilities, with or without replacement	Complete frame, fairly homogeneous population
Stratified	Split into non-overlapping **strata**; sample each independently	Strata internally homogeneous: better precision
Systematic	Random start, then every kth unit; $k = N/n$	Ordered lists, field rows
Cluster	Select clusters at random; **enumerate all** units in them	No unit-level frame; dispersed population
Multistage	Sample in stages (e.g. village → field → plot)	Large-scale surveys such as GCES

Probability sampling allows probability-based inference and measurable sampling error.

5 Non-probability sampling

- **Convenience:** easily accessible units
- **Consecutive:** every accessible eligible unit in turn
- **Quota:** fill subgroup quotas non-randomly
- **Purposive (judgement):** deliberately chosen informative units
- **Snowball:** participants refer others

Useful without a frame or for hard-to-reach groups, but sampling error cannot be measured by probability theory.

6 Survey steps & pioneers

Principal steps: objectives → target population → frame and units → data needed → questionnaire → collection method → **pilot survey** → design and sample size → fieldwork (handle non-response) → edit, tabulate, analyse → report → document.

A.N. Kiaer	Representative Method, 1895
A.L. Bowley	Random sampling and probability-based inference in surveys
P.C. Mahalanobis	Pioneered large-scale sample surveys in India

! Common mistakes

× Calling a convenience sample "random" · A census still has non-sampling errors · SRS draws are independent only **with** replacement · Stratified (sample within every group) ≠ cluster (take whole groups).

? Fill in the blanks

1. Selecting 50 farms out of 1000 gives a sampling fraction of _____. [0.05]
2. To choose 20 plots systematically from 400, the sampling interval k = _____. [20]
3. Splitting a district into irrigated and rainfed areas and sampling from each is _____ sampling. [stratified]
4. Choosing some villages at random and surveying every farmer in them is _____ sampling. [cluster]
5. Wrongly recorded answers can occur even in a census; they are _____ errors. [non-sampling]
6. Interviewing only the farmers who came to a field day is _____ sampling. [convenience]

Source: *Textbook of Agricultural Statistics*, Ch. 12 · Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul · KAU Me-LON

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